Python Course for Physicists (And everyone else!) Chunk 7: Imports

By Jonathan Bollig

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Live Coding: Being Lazy - Let Others Code

- Imports allow you to use code, that others have programmed.
- „Package“ of code: module
- To use a module:

```python
import math
print(math.sqrt(2))
```

```
1.4142135623730951
```

- Direct use of functions:

```python
from math import sqrt
print(sqrt(2))
```

```
1.4142135623730951
```

Live Coding: More options:

- Import multiple functions:

```
from math import sqrt, sin, tan, cos
print(sqrt(2))
print(sin(2))
print(tan(2))
print(cos(2))
```

```
1.4142135623730951
0.9092974268256817
-2.185039863261519
-0.4161468365471424
```

- Import all:

→ Not recommended

```
from math import *
print(atanh(0.5))
```

```
0.5493061443340549
```

- Rename on import

```
from math import log as ln
from math import e
print(ln(e))
```

```
1.0
```

```
import math as m
print(m.log(m.e))
```

```
1.0
```

Built-in Modules → External Modules

- Math is a built-in module in python
- Users upload their modules for use by other people
- For example: numpy ≙ numerical python
- External modules have to be installed in the command line
  → Otherwise:

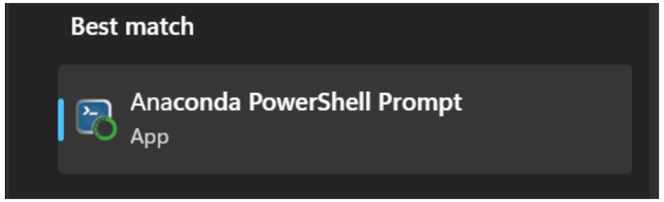
```
import numpy
```

```
---------------------------------------------------------------
ModuleNotFoundError                                   Trac
Cell In[16], line 1
----> 1 import numpy

ModuleNotFoundError: No module named 'numpy'
```

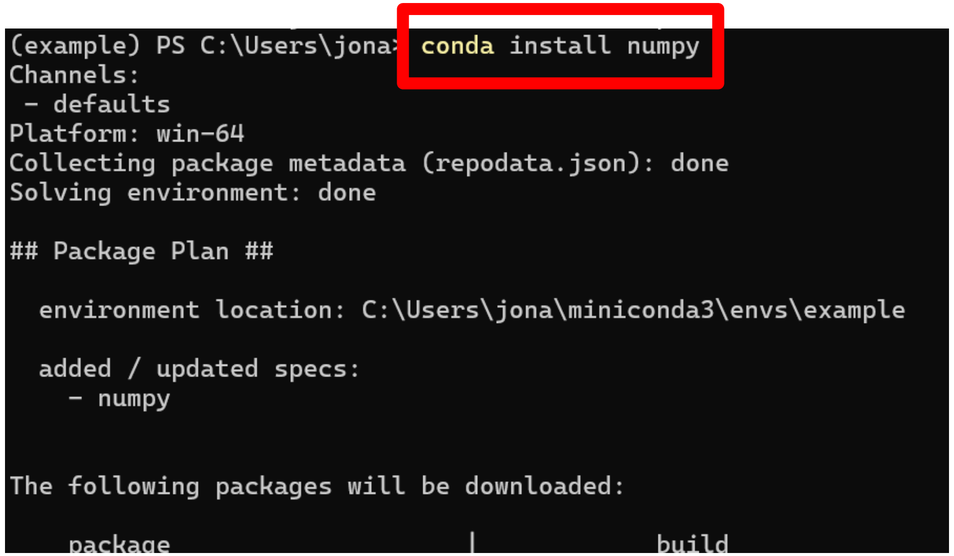
Tutorial: Installing Modules using Conda

- Open “Anaconda Powershell Prompt”

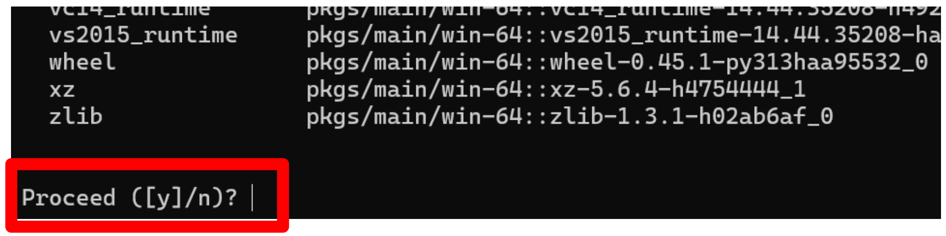


- Enter “conda install numpy”

In your case it will show “base” instead of “example”

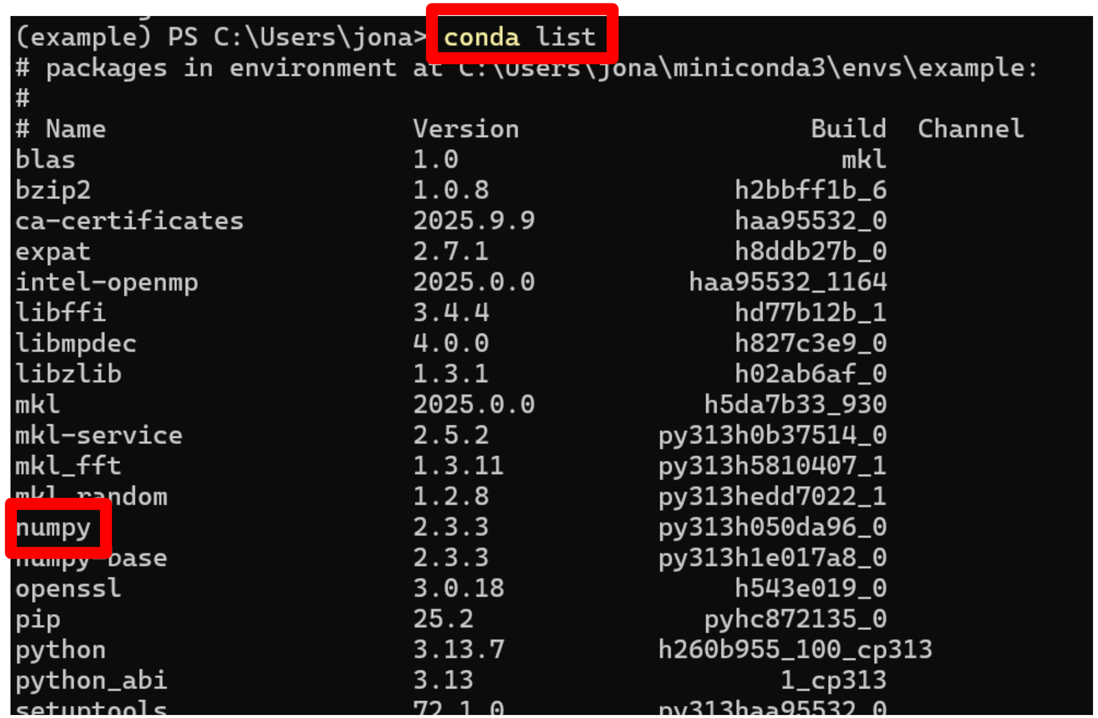


- Type „y“ + Enter to proceed



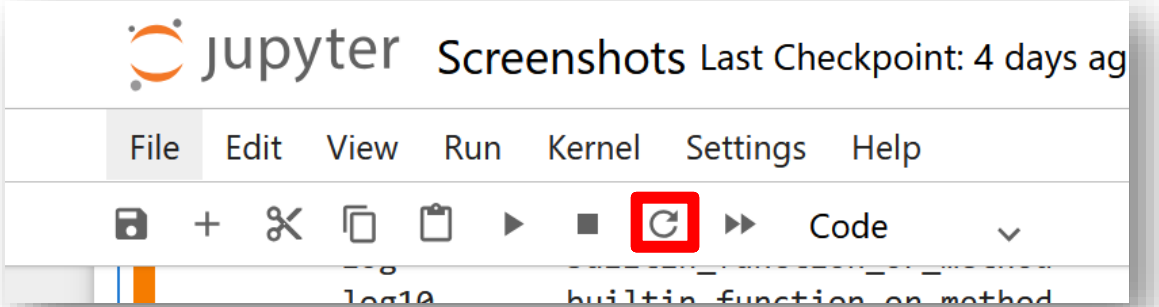
Optional: Check installation with “conda list”

- “conda list” lists all installed packages
- When installing numpy a lot of other packages get installed as well (dependencies)
- Note: “conda” and “pip” do very similar things → mostly interchangeable
(Relevant when googling)



Using installed modules

- Restart kernel:



- Import works now:

```python
import numpy
```

Good Luck with your Tasks ☺

[https://www.reddit.com/r/ProgrammerHumor/comments/o3zjp4/thats_the_evilest_thing_i_can_imagine/]